WORKED SOLUTION

STEP I 1997, Question 13

Probability and Statistics • Conditional Probability

STEP I 1997, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q13

The key idea

Whether Mr Blond should switch on the light depends only on what kind of trap is waiting, so the whole question comes down to one conditional probability: the chance that the trap is an anaconda rather than a bomb, given that there is no canary. The identity of the assassin matters only because it affects which weapon is used. So we combine the two layers (assassin, then weapon) with the law of total probability, condition on "no canary", and then compare the two risks of death.

Solution

Setting up

Let M_i be the event that Mr i set the trap, for $i = 1, 2, 3, 4$, and let A, B and C be the events that the trap is an anaconda, a bomb or a canary. We are given

$$P(M_i) = \frac{i}{10}, \qquad P(A \mid M_i) = \frac{i+1}{10}, \quad P(B \mid M_i) = \frac{i}{10}, \quad P(C \mid M_i) = \frac{9-2i}{10}.$$

These are consistent: $\frac{1+2+3+4}{10} = 1$, and for each i the three weapon probabilities are non-negative and add to $\frac{(i+1)+i+(9-2i)}{10} = 1$. Exactly one assassin and exactly one weapon are involved, so the events $M_i \cap A$, $M_i \cap B$ and $M_i \cap C$ partition all the possibilities.

What the silence tells him

By the law of total probability,

$$P(A) = \sum_{i=1}^{4} P(M_i)P(A \mid M_i) = \frac{1\cdot 2 + 2\cdot 3 + 3\cdot 4 + 4\cdot 5}{100} = \frac{40}{100},$$

$$P(B) = \sum_{i=1}^{4} P(M_i)P(B \mid M_i) = \frac{1 + 4 + 9 + 16}{100} = \frac{30}{100}.$$

So $P(C') = P(A) + P(B) = \frac{70}{100}$. (As a check, $P(C) = \frac{7+10+9+4}{100} = \frac{30}{100}$.)

Ask yourself Do we need to work out which assassin is now most likely?

No. Hearing no singing does change the odds on each assassin, but the risk of death depends only on the weapon. Since $A \subseteq C'$ and $B \subseteq C'$, conditioning on C' simply rescales the two probabilities:

$$P(A \mid C') = \frac{P(A)}{P(C')} = \frac{40}{70} = \frac{4}{7}, \qquad P(B \mid C') = \frac{P(B)}{P(C')} = \frac{30}{70} = \frac{3}{7}.$$

The information about the assassins is already built into these two numbers.

Comparing the two choices

Let K be the event that Mr Blond is killed. Once we condition on C', the trap is an anaconda or a bomb, and his fate depends only on the trap and his choice.

Light on. A bomb kills him with probability $\frac{1}{2}$, and an anaconda kills him with probability $1 - \frac{2}{3} = \frac{1}{3}$. So

$$P(K \mid C', \text{ on}) = \frac{3}{7} \cdot \frac{1}{2} + \frac{4}{7} \cdot \frac{1}{3} = \frac{9}{42} + \frac{8}{42} = \frac{17}{42}.$$

Light off. A bomb never kills him, and an anaconda kills him with probability $\frac{1}{2}$. So

$$P(K \mid C', \text{ off}) = \frac{3}{7} \cdot 0 + \frac{4}{7} \cdot \frac{1}{2} = \frac{2}{7} = \frac{12}{42}.$$

Since $\frac{12}{42} < \frac{17}{42}$, **Mr Blond should enter the flat without switching on the light**. His chance of survival is then $\frac{5}{7}$ (about 71%), compared with $\frac{25}{42}$ (about 60%) if he switches it on.

The reason is that the anaconda is the more likely trap, but it is the bomb that the light really makes worse. Switching on the light lowers the anaconda risk from $\frac{1}{2}$ to $\frac{1}{3}$ but raises the bomb risk from 0 to $\frac{1}{2}$. With probability $\frac{4}{7}$ of an anaconda, the light saves $\frac{4}{7} \cdot \frac{1}{6} = \frac{4}{42}$; with probability $\frac{3}{7}$ of a bomb, it costs $\frac{3}{7} \cdot \frac{1}{2} = \frac{9}{42}$. The cost is larger.

Checks

- Before he listens, the risks would be $\frac{30}{100} \cdot \frac{1}{2} + \frac{40}{100} \cdot \frac{1}{3}$ for the light on and $\frac{40}{100} \cdot \frac{1}{2}$ for the light off. These are the answers above multiplied by $\frac{7}{10}$, as they should be, since the canary contributes nothing to either comparison.
- The light-off strategy is better whenever $p = P(A \mid C')$ satisfies $\frac{1}{2}p < \frac{1}{2}(1-p) + \frac{1}{3}p$, that is, whenever $p < \frac{3}{4}$. Here $p = \frac{4}{7}$, comfortably below that.
- A simulation of 10^6 evenings gives a death rate of about 0.404 with the light on and 0.286 with it off, matching $\frac{17}{42} \approx 0.405$ and $\frac{2}{7} \approx 0.286$.

Common slips

- Forgetting to condition on "no canary" and working with $P(A) = 0.4$ and $P(B) = 0.3$ as if they were the whole story. The advice comes out the same, but the probabilities of death are wrong.
- Mixing up survival and death: the anaconda figure with the light on is a $\frac{2}{3}$ chance of *survival*, so a $\frac{1}{3}$ chance of death.
- Treating the assassins as equally likely, or ignoring the assassin layer and averaging the weapon probabilities.

Beyond STEP

This is a small problem in *decision theory*: we choose the action that minimises expected loss under the posterior distribution, and here the "loss" is simply the probability of death. The observation that only $P(A \mid C')$ matters, and the threshold $p < \frac{3}{4}$ at which the best action switches, are typical of how such decisions are analysed: the data change the posterior, and the posterior decides the action. For more on conditional probability and Bayesian updating, see Blitzstein and Hwang, *Introduction to Probability*.

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